

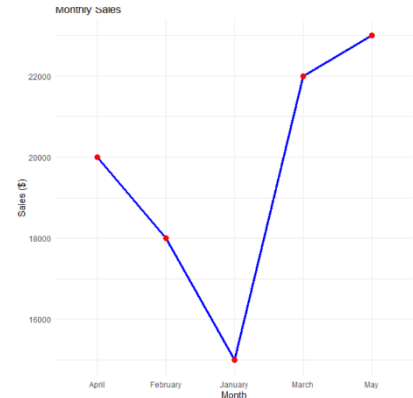
Set 1: Monthly Sales Data

Dataset: Monthly Sales

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

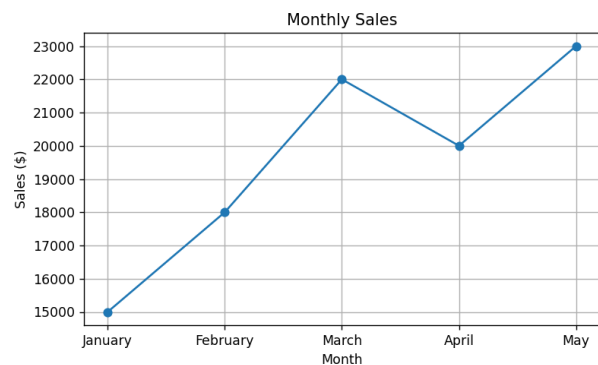
r code –

```
ggplot(sales, aes(x = Month, y = Sales, group = 1)) +  
  geom_line(color = "blue", linewidth = 1.2) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Monthly Sales",  
    x = "Month",  
    y = "Sales ($)"  
  ) +  
  theme_minimal()
```



python code –

```
plt.figure(figsize=(7,4))  
  
plt.plot(sales["Month"],  
        sales["Sales"],  
        marker="o")
```



```
plt.title("Monthly Sales")  
plt.xlabel("Month")  
plt.ylabel("Sales ($)")
```

```
plt.grid(True)
```

```
plt.show()
```

2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

R code –

```
ggplot(product, aes(x = Product, y = Sales, fill = Product)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Top Selling Products",
```

```
x = "Products",
y = "Sales ($)"
) +
theme_minimal() +
theme(legend.position = "none")
```

python code-

```
product = pd.DataFrame({
    "Product":["Product A","Product B","Product C","Product D","Product E"],
    "Sales":[50000,45000,30000,25000,20000]
})
```

```
plt.figure(figsize=(7,4))
```

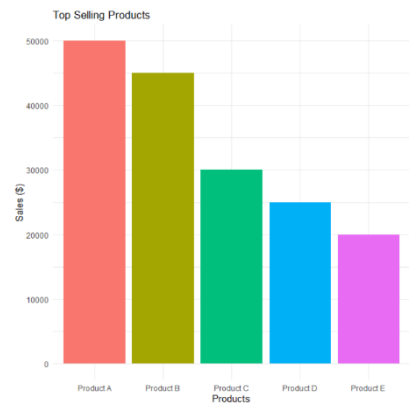
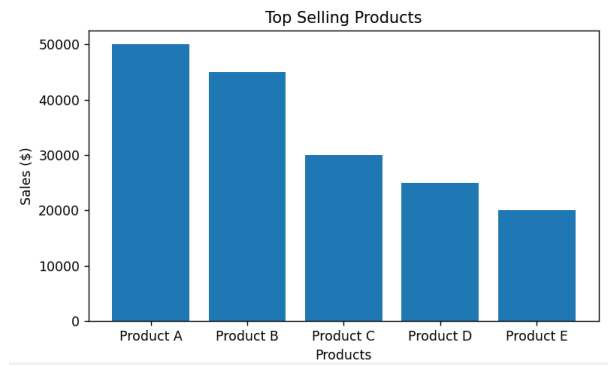
```
plt.bar(product["Product"],
        product["Sales"])
```

```
plt.title("Top Selling Products")
```

```
plt.xlabel("Products")
```

```
plt.ylabel("Sales ($)")
```

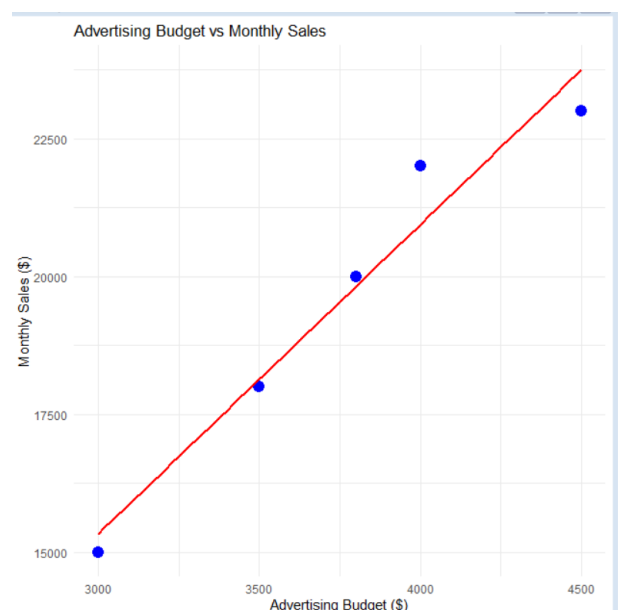
```
plt.show()
```



- Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

R code-

```
ggplot(advertising,
       aes(x = Advertising_Budget,
           y = Sales)) +
  geom_point(size = 4, color = "blue") +
  geom_smooth(method = "lm",
             se = FALSE,
             color = "red") +
  labs(
    title = "Advertising Budget vs Monthly Sales",
```



```

x = "Advertising Budget ($)",
y = "Monthly Sales ($)"
) +
theme_minimal()

python code-
advertising = pd.DataFrame({
    "Advertising Budget":[3000,3500,4000,3800,4500],
    "Sales":[15000,18000,22000,20000,23000]
})

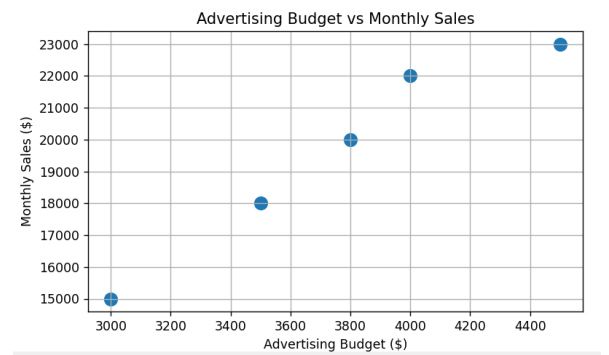
```

```
plt.figure(figsize=(7,4))
```

```

plt.scatter(
    advertising["Advertising Budget"],
    advertising["Sales"],
    s=100
)

```



```

plt.title("Advertising Budget vs Monthly Sales")
plt.xlabel("Advertising Budget ($)")
plt.ylabel("Monthly Sales ($)")

```

```

plt.grid(True)
plt.show()

```

2.Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

R code –

```
install.packages("ggplot2")
install.packages("wordcloud")

library(ggplot2)
library(wordcloud)

customer <- data.frame(
  Customer_ID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction_Score = c(4,5,3,4,5)
)

print(customer)

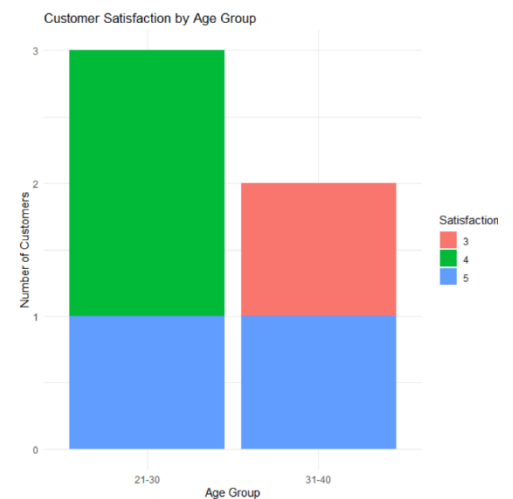
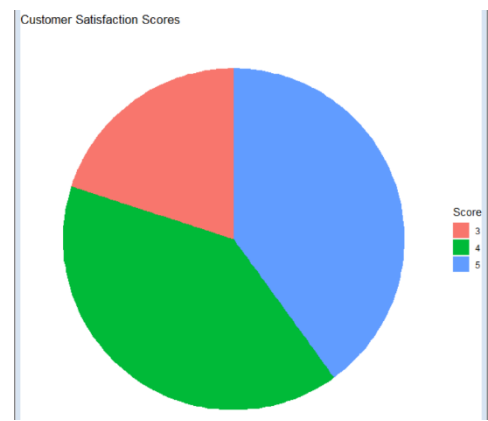
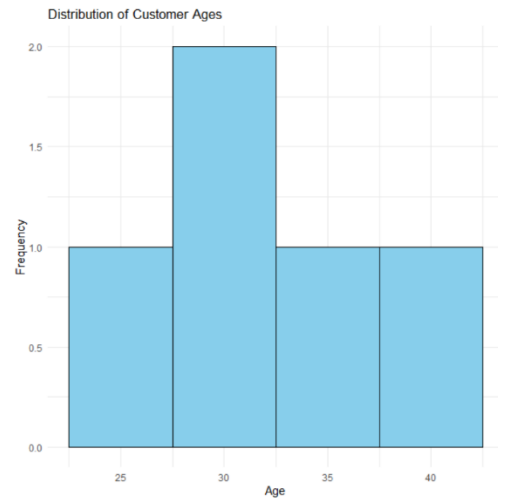
ggplot(customer, aes(x = Age)) +
  geom_histogram(binwidth = 5,
    fill = "skyblue",
    color = "black") +

  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Frequency"
  ) +
  theme_minimal()

score_data <-
as.data.frame(table(customer$Satisfaction_Score))

colnames(score_data) <- c("Score", "Count")

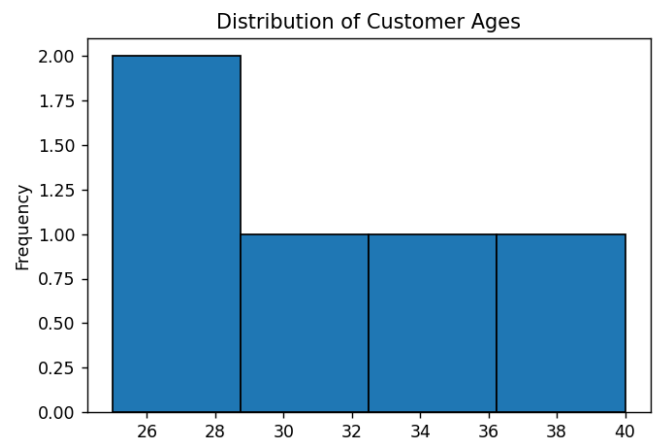
ggplot(score_data,
  aes(x = "", y = Count, fill = Score)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(
```



```

    title = "Customer Satisfaction Scores"
) +
theme_void()
customer$Age_Group <- cut(
  customer$Age,
  breaks = c(20,30,40,50),
  labels = c("21-30","31-40","41-50"),
  include.lowest = TRUE
)
ggplot(customer,
  aes(x = Age_Group,
      fill = factor(Satisfaction_Score))) +
geom_bar() +
labs(
  title = "Customer Satisfaction by Age Group",
  x = "Age Group",
  y = "Number of Customers",
  fill = "Satisfaction"
) +
theme_minimal()

```



python code-

```

import pandas as pd
import matplotlib.pyplot as plt

customer = pd.DataFrame({
    "Customer ID":[1,2,3,4,5],
    "Age":[25,30,35,28,40],
    "Satisfaction Score":[4,5,3,4,5]
})

print(customer)

plt.figure(figsize=(6,4))

```

```

plt.hist(customer["Age"],
         bins=4,
         edgecolor="black")

plt.title("Distribution of Customer Ages")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()

score = customer["Satisfaction Score"].value_counts()

plt.figure(figsize=(5,5))

plt.pie(score,
       labels=score.index,
       autopct="%1.1f%%",
       startangle=90)

plt.title("Customer Satisfaction Scores")

plt.show()

customer["Age Group"] = pd.cut(
    customer["Age"],
    bins=[20,30,40,50],
    labels=["21-30","31-40","41-50"]
)

table = pd.crosstab(
    customer["Age Group"],
    customer["Satisfaction Score"]
)

```

```

table.plot(
    kind="bar",
    stacked=True,
    figsize=(7,5)
)

plt.title("Customer Satisfaction by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Number of Customers")
plt.legend(title="Satisfaction Score")

plt.show()

```

Set 3: Employee Performance Evaluation

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

R code-

```

employee <- data.frame(
  Employee_ID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  Years_of_Service = c(5,3,7,4,2),
  Performance_Score = c(85,92,78,90,76)
)

print(employee)

```

```
ggplot(employee,
  aes(x = Years_of_Service,
    y = Performance_Score,
    color = Department,
    group = Department)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score",
    color = "Department"
  ) + theme_minimal()
```



```
ggplot(employee,
  aes(x = Department,
    fill = Department)) +
  geom_bar() +
  labs(
    title = "Employee Distribution by Department",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

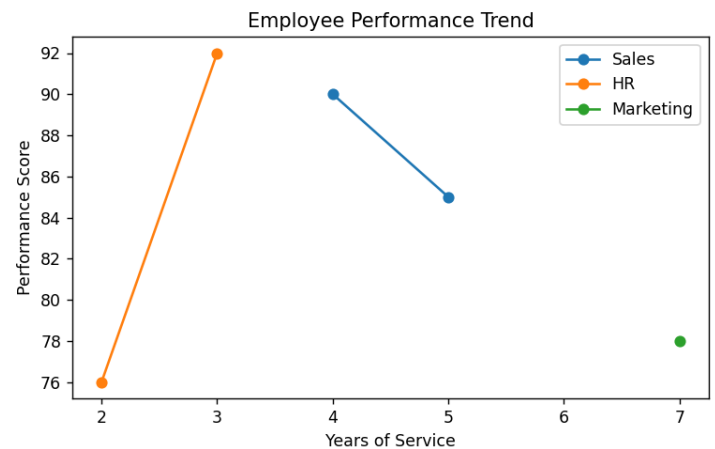
```
ggplot(employee,
  aes(x = Years_of_Service,
    y = Performance_Score,
    color = Department)) +
  geom_point(size = 4) +
  geom_smooth(method = "lm",
    se = FALSE,
    color = "red") +
```



```

labs(
    title = "Years of Service vs Performance Score", x = "Years of Service",
    y = "Performance Score",
    color = "Department"
) +
    theme_minimal()
python code –
import pandas as pd
import matplotlib.pyplot as plt
employee = pd.DataFrame({
    "Employee ID":[1,2,3,4,5],
    "Department":["Sales","HR","Marketing","Sales","HR"],
    "Years of Service":[5,3,7,4,2],
    "Performance Score":[85,92,78,90,76]
})

```



```

print(employee)
plt.figure(figsize=(7,4))

for dept in employee["Department"].unique():
    data = employee[employee["Department"] == dept]
    plt.plot(data["Years of Service"],
             data["Performance Score"],
             marker="o",
             label=dept)

plt.title("Employee Performance Trend")
plt.xlabel("Years of Service")
plt.ylabel("Performance Score")
plt.legend()

plt.show()

```

```
dept_count = employee["Department"].value_counts()
```

```
plt.figure(figsize=(6,4))
```

```
plt.bar(dept_count.index,  
        dept_count.values)
```

```
plt.title("Employee Distribution by Department")
```

```
plt.xlabel("Department")
```

```
plt.ylabel("Number of Employees")
```

```
plt.show()
```

```
plt.figure(figsize=(7,4))
```

```
plt.scatter(employee["Years of Service"],  
            employee["Performance Score"],  
            s=100)
```

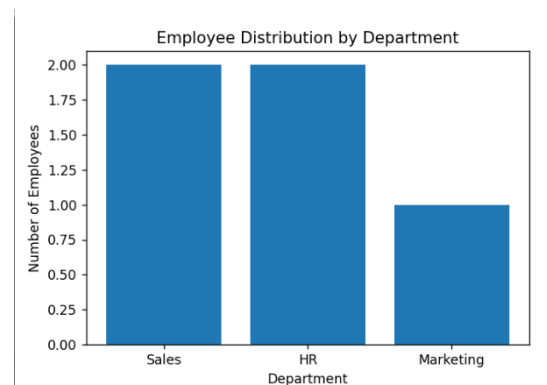
```
plt.title("Years of Service vs Performance Score")
```

```
plt.xlabel("Years of Service")
```

```
plt.ylabel("Performance Score")
```

```
plt.grid(True)
```

```
plt.show()
```



Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200

Product ID	Product Name	Quantity Available
5	Product E	220

R code –

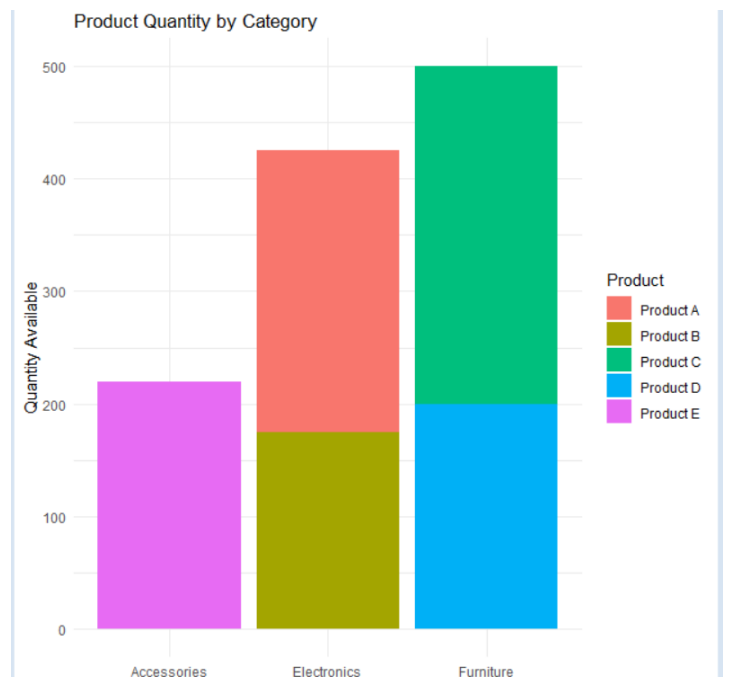
```
inventory <- data.frame(
  Product_ID = c(1,2,3,4,5),
  Product_Name = c("Product A","Product B","Product C","Product D","Product E"),
  Quantity = c(250,175,300,200,220)
)
```

```
print(inventory)
```

```
ggplot(inventory,
  aes(x = Product_Name,
    y = Quantity,
    fill = Product_Name)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Product Inventory",
    x = "Product Name",
    y = "Quantity Available"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
inventory$Category <- c("Electronics",
  "Electronics",
  "Furniture",
  "Furniture",
  "Accessories")
```

```
print(inventory)
```

```
ggplot(inventory,
  aes(x = Category,
```



```

        y = Quantity,
        fill = Product_Name)) +
geom_bar(stat = "identity") +
labs(
  title = "Product Quantity by Category",
  x = "Category",
  y = "Quantity Available",
  fill = "Product"
) +
theme_minimal()

```

python code-

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
inventory = pd.DataFrame({
```

```
    "Product ID":[1,2,3,4,5],
```

```
    "Product Name":["Product A",  

"Product B","Product C",  

"Product D","Product E"],
```

```
    "Quantity":[250,175,300,200,220]
```

```
})
```

```
print(inventory)
```

```
plt.figure(figsize=(7,4))
```

```
plt.bar(inventory["Product Name"],  

inventory["Quantity"])
```

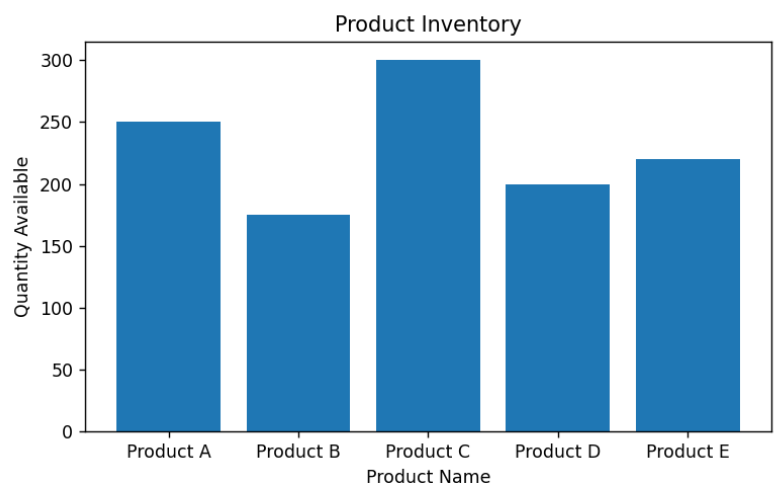
```
plt.title("Product Inventory")
```

```
plt.xlabel("Product Name")
```

```
plt.ylabel("Quantity Available")
```

```
plt.show()
```

```
inventory["Category"] = [
```



```

    "Electronics",
    "Electronics",
    "Furniture",
    "Furniture",
    "Accessories"
]

```

```

pivot = inventory.pivot_table(
    index="Category",
    columns="Product Name",
    values="Quantity",
    fill_value=0
)

```

```

pivot.plot(
    kind="bar",
    stacked=True,
    figsize=(8,5)
)

```

```

plt.title("Product Quantity by
Category")
plt.xlabel("Category")
plt.ylabel("Quantity Available")

```

```

plt.show()
inventory["Price"] = [120,150,100,180,130]

```

```

plt.figure(figsize=(7,4))

```

```

plt.scatter(
    inventory["Price"],
    inventory["Quantity"],

```



```
s=100  
)
```

```
plt.title("Product Price vs Quantity")  
plt.xlabel("Price ($)")  
plt.ylabel("Quantity Available")
```

```
plt.grid(True)  
plt.show()
```

Set 5: Website Analytics

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

R code-

```
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03",  
    "2023-01-04","2023-01-05")),  
  Page_Views = c(1500,1600,1400,1650,1800),  
  CTR = c(2.3,2.7,2.0,2.4,2.6)  
)
```

```
print(traffic)  
ggplot(traffic,  
  aes(x = Date,  
    y = Page_Views,  
    group = 1)) +  
  geom_line(color = "blue", linewidth = 1.2) +
```

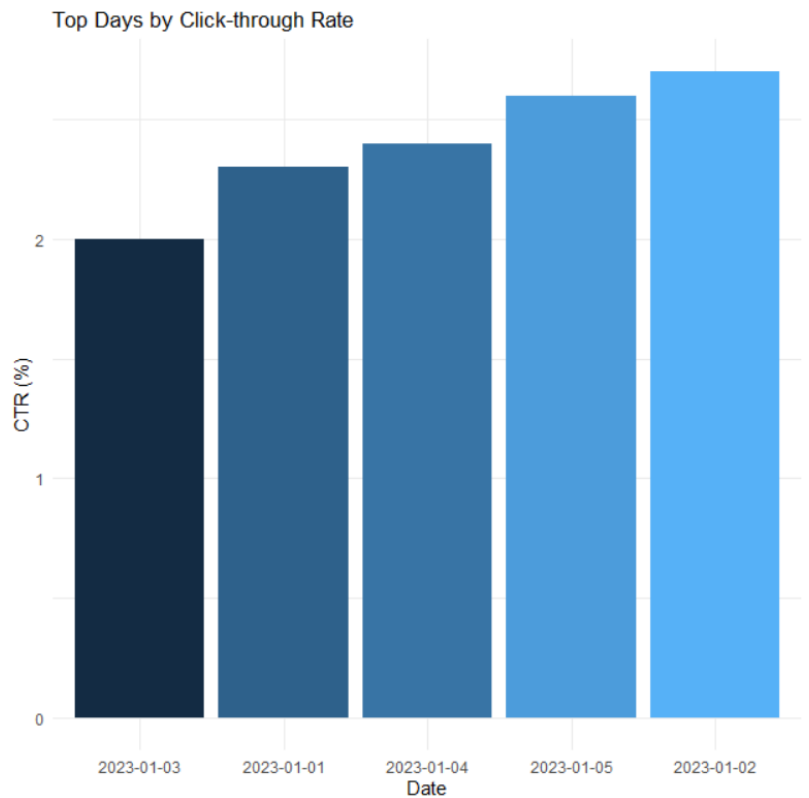
```
geom_point(color = "red", size = 3) +
labs( title = "Daily Page Views",
x = "Date",
y = "Page Views"
) +
theme_minimal()
top_ctr <- traffic[order(-traffic$CTR), ]
```

```
ggplot(top_ctr,
aes(x = reorder(as.character(Date),
CTR),
y = CTR,
fill = CTR)) +
geom_bar(stat = "identity") +
labs(
title = "Top Days by Click-through
Rate",
x = "Date",
y = "CTR (%)"
) +
theme_minimal() +
theme(legend.position = "none")
```

python code –

```
import pandas as pd
import matplotlib.pyplot as plt
traffic = pd.DataFrame({
    "Date":["2023-01-01","2023-01-02","2023-01-03",
        "2023-01-04","2023-01-05"],
    "Page Views":[1500,1600,1400,1650,1800],
    "CTR":[2.3,2.7,2.0,2.4,2.6]
})

traffic["Date"] = pd.to_datetime(traffic["Date"])
```



```
print(traffic)

plt.figure(figsize=(7,4))

plt.plot(
    traffic["Date"],
    traffic["Page Views"],
    marker="o"
)

plt.title("Daily Page Views")
plt.xlabel("Date")
plt.ylabel("Page Views")

plt.grid(True)
plt.show()

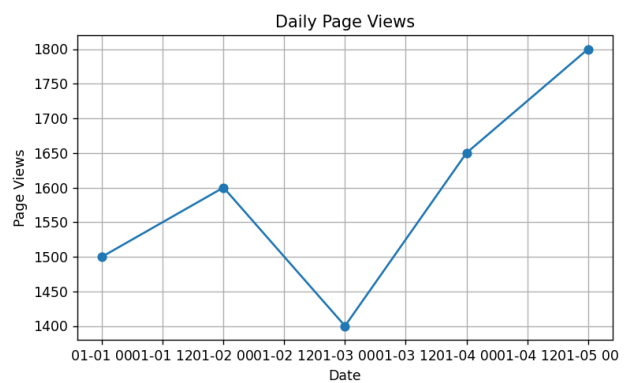
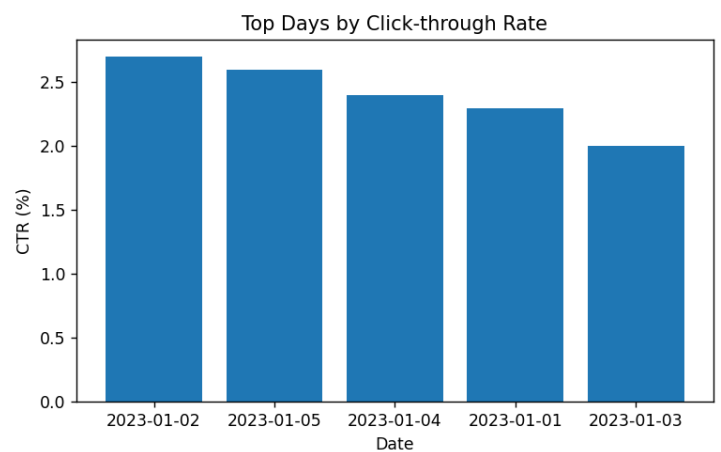
top_ctr = traffic.sort_values(
    by="CTR",
```

```
    ascending=False
```

```
)
```

```
plt.figure(figsize=(7,4))
```

```
plt.bar(
```




```
top_ctr["Date"].dt.strftime("%Y-%m-%d"),  
top_ctr["CTR"]  
)
```

```
plt.title("Top Days by Click-through Rate")  
plt.xlabel("Date")  
plt.ylabel("CTR (%)")  
plt.show()
```